

JEAN NAUDE

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SUMMARY

Senior Machine Learning Engineer with 5 years of experience delivering production-level ML models for a major retailer. Former Co-founder of a non-profit that developed open-source epidemiology simulation software used by the World Health Organization.

My strength lies in rapidly deploying observable, impactful models that drive key business decisions in their organizations. I enjoy taking on difficult problems and developing solutions at a rapid pace whilst maintaining measurable and observable improvement in automation, optimization and model accuracy. I enjoy working with hardworking people who want to get results fast, can communicate effectively, and are open to feedback so that decisions can be made and improvements can be implemented.

I am extremely passionate and competent in building AI agents and workflows that can automate complex tasks and improve productivity. I have experience building and deploying AI agents using Claude Code, OpenCode, MCP tool setup, A2A (Agent to Agent), ACP, Skills, agent orchestration, testing frameworks, playwright, evals, Claude Code agent orchestration pipelines, and Generative AI workflows and orchestration. I am always looking for new ways to improve my skills and stay up-to-date with the latest technologies.

Strong hands-on MLOps experience: designing GitLab CI/CD pipelines with clear build/test/deploy/release stages and integrated vulnerability scanning (dependencies, containers, and IaC), implemented automated retraining and deployment pipelines, and monitoring on Grafana/cloud monitoring for data drift with structured logging, alerting, and model explainability/score attribution using SHAP (Shapley values). Experienced in deploying and scaling Kubernetes clusters, and selecting pragmatic serving patterns on GCP (Cloud Run/Vertex AI/Pub Sub) and AWS (Bedrock/Sagemaker) depending on latency, cost, and operational overhead.

In other words, I get shit done!

EXPERIENCE

Machine Learning Engineer — Spatialedge

Jun 2021 – Present

- Served as Technical Lead Engineer on two enterprise data applications providing optimized results in retail and financial sectors
 - Mentored junior engineers and provided technical guidance to engineering teams while directing project execution and establishing the technical foundation for both projects
 - Productionized and contributed to 8 machine learning models including Lead Time Forecasting, Demand Forecasting, and Interbranch Transfers Optimization in the Retail sector. And ATM replenishment forecasting and optimization in the Financial Sector
 - Contributed development to various credit scoring models in the Financial sector, and mentored 5 junior engineers on software engineering best practices and approaches
 - Developed Workforce Planning Optimization and Cash Replenishment Forecasting enterprise-grade data applications
 - Implemented CI/CD pipelines using GitLab and GCP with Terraform for infrastructure automation
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SKILLS

Tools: Claude Code, OpenCode, MCP tool setup, agent orchestration, testing frameworks, playwright, Claude Code agent orchestration pipelines, Generative AI workflows and orchestration

Languages & Frameworks: Python, C++, SQL, JavaScript, TypeScript, Ruby, Bash, C

ML & Data Science: PyTorch, Scikit-Learn, TensorFlow, Time Series Forecasting, Machine Learning, Computer Vision

Infrastructure & DevOps: GCP, Azure, Docker, Kubernetes, Terraform, CI/CD, Airflow, Dagster, Helm, Git, Linux

Databases & Systems: PostgreSQL, BigQuery, Redis, CUDA, GraphQL, Crow API

EDUCATION

ESIEE PARIS — Master's of Science in Computer Science

2019-2021

Pearson Institute — Bachelor's of Science in Information Technology

2015-2018

PROJECTS

VISS (Viral Infection Simulation Subsystem)

github.com/twigglits/viss

- Engineered first open-source Viral Infection Simulation Subsystem providing valuable insights to the World Health Organization
- Implemented backend using Redis DB, C++, and Crow API with CUDA acceleration for high-performance epidemiological modeling

- Developed modular frontend using React and TypeScript with CSS Tailwind for interactive simulation visualization

Spatialedge Analytics DF Auditor

pypi.org/project/spatialedge-analytics-dfauditor

- Created DataFrame auditing and validation tools for machine learning pipeline quality assurance
- Published Python package to PyPI for broader community use in data science workflows

HACKATHON PROJECTS

AI Kernel Killswitch — LLM Safety & Containment

github.com/twigglits/AI_kernel_killswitch

- Built a last-resort containment mechanism for self-hosted LLMs: a harness-gated killswitch that verifies an authenticated AES-256-GCM(KILL) payload embedded in a prompt, sets a fail-closed fuse, and fires two independent, irreversible erasure paths (in-worker vLLM weight scramble and LUKS crypto-shred)
- Designed replay-protected, key-off-disk detonation with an air-gapped golden master preserved for recovery and forensics
- Ran a companion research track: LoRA-trained sleeper-agent trojan on TinyLlama-1.1B (perfect recall, zero leakage), a linear activation detector at 1.0 held-out accuracy, and an inference-time monitor — delivering a first-class negative result that single-direction ablation does not remove baked-in backdoors

CERTIFICATIONS

Python for Data Science and Machine Learning Bootcamp — 2025

<https://www.udemy.com/certificate/UC-1fcc8e47-b153-45b2-b43b-7c117e1fd2db/>

Time Series Analysis, Forecasting and Machine Learning — 2025

<https://www.udemy.com/certificate/UC-a567a1fa-c83a-43d9-82ff-df5cdacb4d5e/>

Data Science and Machine Learning Bootcamp with R — 2019

<https://www.udemy.com/certificate/UC-65MDQ02B/>

React - The Complete Guide 2025 (incl. Next.js, Redux) — 2025

<https://www.udemy.com/certificate/UC-acdec4cd-ae82-4216-9cd3-212d2e0a63ee/>

Calculus 1 — 2019

<https://www.udemy.com/certificate/UC-37U23IL1/>

HOBBIES

Exploring number theory — experimental work on the open **Erdős–Straus conjecture**, building four independent solution-counting engines (C, Rust, CUDA, Python) agreeing byte-for-byte to compute the largest per-prime dataset I am aware of (~278k hard-class primes up to 2×10^9), confirming empirical lognormal laws via blind minima prediction, machine-verifying proof-attempt theorems, and extending the problem to the negative integers. github.com/twigglits/erdos-strauss-conjecture--expansion